TEJASVI A JAVAGAL

Aspiring Data Engineer | Python | SQL | ETL Pipelines | Data Pipelines

Bengaluru, Karnataka, IN | +91 8971964665 | tejasvijavagal@gmail.com | github.com/javagaltejasvi46 | linkedin.com/in/tejasvijavagal

CAREER OBJECTIVE

Final-year B.Tech (Information Science) student specializing in Python and SQL, with hands-on experience designing end-to-end ETL pipelines and data-intensive backend systems as an aspiring Data Engineer. Skilled at building ingestion, transformation, and storage workflows across structured and unstructured data using PostgreSQL and SQLite. Comfortable with schema design, query optimization, and containerized deployment (Docker), backed by a track record of shipping reliable data pipelines end to end. Looking to bring this foundation to a role building scalable, production-grade data platforms.

SKILLS

Languages: Python, SQL, JavaScript, C++, C
Data & ETL: ETL Pipelines, Data Pipelines, Data Ingestion & Transformation, Feature Engineering, REST APIs, Batch Processing
Databases: PostgreSQL, MySQL, SQLite3, MongoDB (NoSQL) — schema design, query optimization
Frameworks & Libraries: FastAPI, Flask, Django, SQLAlchemy, Pydantic, PyTorch, TensorFlow, LangChain, Pandas, OpenCV
AI / ML: Deep Learning, NLP, LLMs, Model Training & Inference, TabNet, GNN, XGBoost
Tools: Git, GitHub, Docker, Vite, React, TypeScript, WebSockets

PROJECTS

Garud Drishti – Landslide Early-Warning Data Pipeline (SIH) | github.com/Ysh0910/garud-drishti *Aug 2026*

Python, PostgreSQL/PostGIS, XGBoost, FastAPI — Geospatial Data Engineering

- **Multi-source data ingestion:** Built the ingestion and feature-engineering pipeline consolidating heterogeneous geospatial sources — GSI/ISRO historical landslide records, IMD/GPM rainfall data, and SMAP soil-moisture data — into structured static and dynamic feature stores.
- **Schema design:** Designed a PostgreSQL/PostGIS schema to store and query terrain, geology, and time-series risk features efficiently, supporting fast lookups for a GIS-backed risk engine.
- **ETL & transformation:** Transformed raw satellite and sensor data into model-ready features (elevation, slope, aspect, land cover, rainfall/soil-moisture aggregates), feeding a two-stage risk-scoring pipeline across 6h/24h/48h/72h horizons.
- **Data quality:** Prioritized clean, well-structured, explainable feature pipelines over ad hoc transformations, balancing pipeline maintainability with downstream model accuracy for authority-facing risk outputs.

SANJAYA SmartRoute AI – Traffic ETL Pipeline & Analysis | github.com/javagaltejasvi46/mini-project *Mar 2026*

Python, FastAPI, PostgreSQL, TabNet, GNN, MongoDB

- **ETL pipeline:** Designed a multi-source data ingestion and transformation pipeline using FastAPI, extracting raw traffic feeds, transforming with TabNet (deep learning), and loading structured outputs into MongoDB — processing 10K+ records per run.
- **SQL & schema design:** Modelled relational data in PostgreSQL for historical traffic snapshots; wrote optimized SQL queries for aggregation, time-series extraction, and route performance comparison across 50+ road segments.
- **Automated retraining:** Built a feedback loop that retrains the model every 100 samples, sustaining 75–85% prediction accuracy with zero manual intervention.

SwarSanchar – Audio Data ETL Pipeline | github.com/javagaltejasvi46/SwarSanchar.git *Jan 2026*

FastAPI, Demucs, PostgreSQL, TypeScript

- **ETL for unstructured data:** Built an end-to-end ETL pipeline for audio — Extract (yt-dlp acquisition), Transform (pitch modification, format conversion via Demucs), Load (multi-format export) — processing 30+ tracks per batch with ~70% reduction in manual effort.
- **SQL data store:** Designed a PostgreSQL schema to track job status, processing state, and output file paths; wrote SQL queries for job queue management and history retrieval.
- **Scalable inference:** Integrated Demucs as a backend ML inference service, optimizing batch throughput for concurrent audio transformation requests.

EDUCATION

Reva University — B.Tech, Information Science *Jul 2023 – Jul 2027*

CGPA: 7.9 / 10.0 | Relevant Coursework: Data Structures & Algorithms, Database Management Systems, Operating Systems

CERTIFICATIONS

Python for Data Science, AI & Development — Udemy View Certificate
DSA in Python — Udemy View Certificate